National University of Computer and Emerging Sciences
Karachi Main Campus

Machine Learning (AI3002)

Final Term

Date: December 26, 2024

Course Instructors

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Total Time (Hrs): 3

Total Marks: 100

Total Questions: 5

_____________ Roll Number   _____________ Section   ________________________ Student Signature

Attempt all the questions.

CLO # 3: Understand the core theoretical concepts serving as foundations of ML algorithms. **[2.5x8=20 Marks]**

Q1. Answer the following questions briefly.

a. How do biological neurons differ from artificial neurons in terms of structure and function?
b. Given the true labels, we can use our intuition for weight update, so why need a learning algorithm?
c. Why do we desire linear separability for machine learning models?
d. What is the role of hidden layers in a multilayer neural network, and how do they contribute to the model's capacity to learn?
e. How does the chain rule simplify the computation of gradients, and why is this simplification crucial for efficient training?
f. In Naive Bayes, why is it important for features to be conditionally independent?
g. How do you decide how many principal components to retain in PCA?
h. Explain the concept of linkage criteria in hierarchical clustering and how it affects the clustering result.

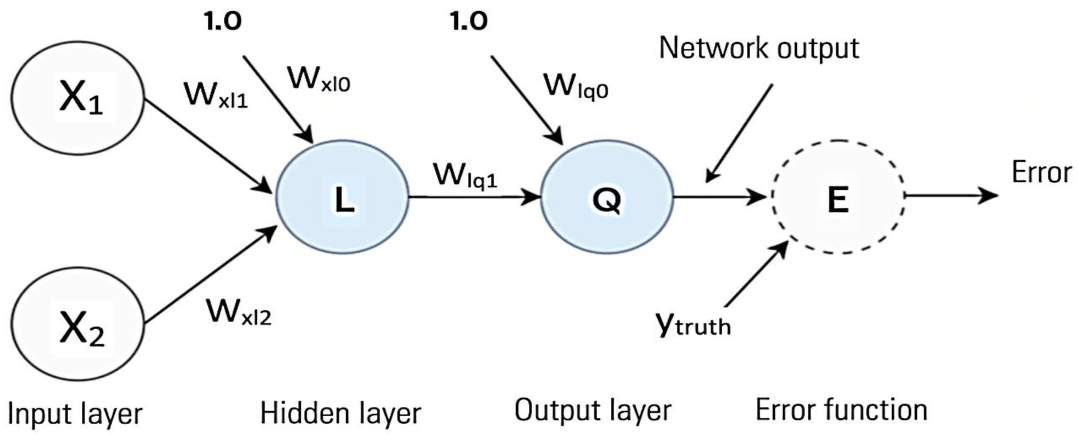
CLO # 1: Understand and recognize a machine learning problem. **[20 Marks]**

Q2. Given that the perceptron is being applied to the PIMA diabetes dataset for binary classification. Based on two features age and glucose, and their corresponding labels of three female patients, find out optimal weights using the perceptron learning algorithm that performs correct classification given that the initial weights are $w_0 = 0.51,\ w_1 = -3.13,\ w_2 = 2.94,\ X_0 = 1, and\ \eta = 0.01$.

Age	Glucose	Label
33	317	1
30	110	-1
41	237	1

CLO # 4: Understand the trade-offs among model complexity, data size, and model performance for different algorithms. **[10+15=25 Marks]**

Q3 a. Briefly explain how backpropagation will be applied to the below network using pseudo code or algorithm.



Q3 b. The architecture in part a is given $X = \begin{bmatrix} -0.7 \\ 0.2 \end{bmatrix}$, $W = \begin{bmatrix} 0.2 \\ 0.5 \\ -0.2 \\ -0.1 \\ 0.4 \end{bmatrix}$, Y_{truth} = -1, and $\eta = 0.4$.

Using backpropagation algorithm find the updated weights.

CLO # 5: Apply and interpret probabilistic machine learning methods on real-world datasets. **[15 Marks]**

Q4. A manufacturing plant produces various products, and quality control is crucial for ensuring that only products meeting the standards are shipped out. To automate the quality assessment process, the plant employs a Naive Bayes classifier that uses several features extracted from product inspection data. The goal is to predict whether a product is High Quality (HQ) or Low Quality (LQ). Below is a dataset from a recent batch of products produced in the plant. Each row represents a product, its associated features, and whether the product was High Quality (HQ) or Low Quality (LQ).

Material Type	**Temperature**	**Surface Defects**	**Packaging**	**Production Speed**	**Quality**
Steel (S)	High (H)	Few (F)	Premium (P)	Slow (S)	HQ
Aluminum (A)	Medium (M)	None (N)	Standard (S)	Fast (F)	LQ
Plastic (P)	Low (L)	Many (M)	Standard (S)	Normal (N)	LQ
Steel (S)	High (H)	Few (F)	Premium (P)	Slow (S)	HQ
Aluminum (A)	Low (L)	None (N)	Standard (S)	Slow (S)	HQ
Plastic (P)	Medium (M)	Few (F)	Premium (P)	Fast (F)	LQ
Steel (S)	Low (L)	Many (M)	Standard (S)	Normal (N)	LQ
Aluminum (A)	Medium (M)	Few (F)	Premium (P)	Normal (N)	HQ
Plastic (P)	High (H)	Few (F)	Standard (S)	Fast (F)	LQ
Steel (S)	Medium (M)	None (N)	Premium (P)	Slow (S)	HQ

Using Naive Bayes, classify a new product with the following attributes:

Material Type	**Temperature**	**Surface Defects**	**Packaging**	**Production Speed**	**Quality**
Plastic (P)	High (L)	None (N)	Premium (P)	Slow (F)	?
Steel (S)	Low (H)	Few (F)	Standard (S)	Normal (N)	?

CLO # 2: Formulate and execute solutions to the machine learning problems. **[15+5=20 Marks]**

Q5. The table below represents the pairwise distance values between seven regional offices of a multinational company.

	RO1	**RO2**	**RO3**	**RO4**	**RO5**	**RO6**	**RO7**
RO1	0	936	417	681	515	765	849
RO2	936	0	559	344	255	690	702
RO3	417	559	0	702	640	600	780
RO4	681	344	702	0	995	725	930
RO5	515	255	640	995	0	559	570
RO6	765	690	600	725	559	0	702
RO7	849	702	780	930	570	702	0

a. Perform Agglomerative Clustering using the single-linkage method to group the regional offices based on the distances. Show the updated distance table after each merge until all offices are grouped into one cluster.
b. Visualize the process using a dendrogram.